In distributed systems, including microservices and cloud native applications, the most-used strategy for exchanging information about an authenticated user and their authorization is through tokens.

JSON Web Token (JWT) is an industry-standard for representing *claims* to be transferred between two parties. It's a widely used format for propagating information about an authenticated user and their permissions securely among different parties in a distributed system. A JWT is not used by itself, but it's included in a larger structure, the JSON Web Signature (JWS), which ensures the integrity of the claims by digitally signing the JWT object.

A digitally signed JWT (JWS) is a string composed of three parts encoded in Base64 and separated by a dot (`.`) character:

```
<header>.<payload>.<signature>
```

NOTE For debugging purposes, you can use the tool available on https://jwt.io to encode and decode tokens.

As you saw, a digitally signed JWT has three parts:

- *Header*—A JSON object (called *JOSE Header*) containing information about the cryptographic operations performed on the payload. The operations follow the standards from the Javascript Object Signing and Encryption (JOSE) framework. A decoded header looks like this:

```
{
  "alg": "HS256",     ❶
  "typ": "JWT"        ❷
}
```

❶ The algorithm used to sign the token digitally

❷ The type of token